VERITARIFF

THE TRADE COMPLIANCE OS

Happy Path Workflow — Developer Functional Specification

Steel Export Corridor: United Kingdom → Federal Republic of Germany

HS Chapters 72 & 73 | AEO-in-a-Box Compliance Automation | v3.0

Classification	INTERNAL — CO-FOUNDER WORKING DOCUMENT
Corridor	United Kingdom → Federal Republic of Germany
Commodity	Iron & Steel Products, HS Chapters 72 & 73
Version	v3.0 — Aligned to Founder Legal Analysis Workflow
Date	March 2026
Product	Veritariff — veritariffai.co
Legal Frameworks	UK Trade Tariff · UK-EU TCA · EU CBAM Reg 2023/956 · UK Russia Sanctions SI 2019/855

💡 This specification is structured to mirror the exact decision-tree logic analysed by the founder from primary UK and EU law. Each step maps to a user-facing screen or automated engine action within the Veritariff platform.

STEP 1 · CLASSIFY THE COMMODITY CODE

First gate of the Happy Path — is the HS code already known?

The commodity code is the foundation of every compliance decision downstream. Veritariff checks whether the code was already determined in the calculator flow before triggering the full classification engine.

? Has the commodity code already been classified (e.g. from the calculator flow)?

✅ **YES** → Save it to the shipment record and display a 'Classified ✓' tick in the workflow timeline. Proceed to Step 2.

❌ **NO** → Trigger the Steel Classification Engine (Step 1.1 → 1.2 below).

Why the Commodity Code Matters

⚠️ The 10-digit commodity code is the single most important data point in any customs declaration. Everything downstream — duty rates, VAT, licences, preferences, CBAM, sanctions flags — depends on it being correct.

The UK Trade Tariff tool (and Veritariff's classification engine) uses the code to determine:

- The rate of Customs Duty and import VAT
- Applicable taxes and excise
- Preferential rates under trade agreements (e.g. UK-EU TCA)
- Whether an import/export licence is required
- Whether goods are covered by agricultural policy
- Whether anti-dumping duties apply
- Whether UK safeguarding measures apply
- Whether tariff quotas (TRQs) apply

ℹ️ Only the first 6 digits of a commodity code are internationally standardised (the HS code). Digits 7–10 are country-specific and must be validated against the destination country's database — for Germany, this means cross-referencing with the EU TARIC / CN system.

Step 1.1 — Chemical Composition Gate

STEP 1.1 — Chemical Composition — The Base Material Gate

LOGIC

```
// System asks: 'What is the exact composition of the metal?'

IF   material == 'waste_or_scrap'                        → HEADING 7204
IF   material == 'granules_or_powders'                   → HEADING 7205
IF   material == 'direct_reduction_sponge_iron'          → HEADING 7203
IF   carbon_pct > 2.0                                    → HEADING 7201  [Pig Iron]
IF   manganese_pct > 6.0                                 → HEADING 7201
[Spiegeleisen]
```

```
IF   composition == ferro_alloy                          → HEADING 7202

IF   carbon_pct <= 1.2  AND  chromium_pct >= 10.5
     → class = 'STAINLESS_STEEL'         → proceed to Step 1.2A

IF   (aluminium_pct >= 0.3 OR manganese_pct >= 1.65 OR
      other_alloy_minimums_met)  AND  class != 'STAINLESS'
     → class = 'OTHER_ALLOY_STEEL'       → proceed to Step 1.2B

ELSE → class = 'NON_ALLOY_IRON_STEEL'   → proceed to Step 1.2C
```

DATA INPUTS

Carbon %	Percentage carbon content of the steel
Chromium %	Percentage chromium (key for stainless classification)
Manganese %	Percentage manganese
Aluminium %	Percentage aluminium
Material Type	Scrap / granules / sponge iron / pig iron / ferro-alloy / other

USER PROMPT

UI screen: 'What is the exact composition of the metal?' with guided input fields for each chemical element percentage, plus a dropdown for material type (scrap, granules, etc.).

REG REF

UK Trade Tariff — Chapter 72 Notes | EU Combined Nomenclature 2024 Section XV Chapter 72 Legal Notes

Step 1.2 — Physical Form Assignment

Once chemical class is set, the system asks: 'What shape or form is the metal currently in?' The 4-digit heading is assigned from the table below.

Step 1.2A — If Class = STAINLESS STEEL

```
IF  form == 'ingots' OR 'primary_forms' OR 'semi_finished'  →  HEADING 7218
IF  form == 'flat_rolled' AND width_mm >= 600  →  HEADING 7219
IF  form == 'flat_rolled' AND width_mm < 600  →  HEADING 7220
IF  form == 'bars_rods_hot_rolled_irregular_coils'  →  HEADING 7221
IF  form == 'bars_rods_other' OR 'angles' OR 'shapes' OR 'sections'  →  HEADING 7222
IF  form == 'wire'  →  HEADING 7223
```

Step 1.2B — If Class = OTHER ALLOY STEEL

```
IF  form == 'ingots' OR 'primary_forms' OR 'semi_finished'  →  HEADING 7224
IF  form == 'flat_rolled' AND width_mm >= 600  →  HEADING 7225
IF  form == 'flat_rolled' AND width_mm < 600  →  HEADING 7226
IF  form == 'bars_rods_hot_rolled_irregular_coils'  →  HEADING 7227
IF  form == 'bars_rods_other' OR 'angles' OR 'shapes' OR 'sections' OR
'hollow_drill_bars'  →  HEADING 7228
```

Step 1.2C — If Class = NON-ALLOY IRON / STEEL

```
IF form == 'ingots' OR 'primary_forms' (blocks, lumps) → HEADING 7206
IF form == 'semi_finished' (solid section, cont. cast, blooms, billets) → HEADING 7207
IF form == 'flat_rolled' AND width_mm >= 600 → HEADING 7208, 7209, or 7210
IF form == 'flat_rolled' AND width_mm < 600 → HEADING 7211 or 7212
IF form == 'bars_rods_hot_rolled_irregular_coils' → HEADING 7213, 7214, or 7215
IF form == 'angles_shapes' (L-angles, omega profiles) → HEADING 7216
IF form == 'wire' → HEADING 7217
```

Step 1.3 — Post-Classification Output

After classification, Veritariff stores and displays the following to the user:

10-Digit Commodity Code	Full code validated against UK Trade Tariff API
HS Chapter / Heading	4-digit heading (e.g. 7208) and chapter (72)
Supplementary Unit	Additional measurement unit for customs declaration (e.g. tonnes, number of items) — required alongside weight in kg
Classification Confidence	High / Medium / Low based on user inputs provided
Licence Check	Auto-flag if a licence is required for this code
Safeguard / TRQ Flag	Auto-flag if safeguarding measures or tariff quotas apply

Step 1.4 — API & Attribution

⚠️ Veritariff uses the UK Government Trade Tariff API. The Open Government Licence v3.0 requires the following attribution to be displayed in the application.

API endpoint:

https://api.trade-tariff.service.gov.uk — Trade Tariff API

Reference documentation:

gov.uk — Finding commodity codes for imports or exports

🚫 MANDATORY ATTRIBUTION — Must be displayed somewhere in the Veritariff application: 'Contains public sector information licensed under the Open Government Licence v3.0.' Include a link to the licence. Failure to comply automatically terminates your right to use the data under the OGL.

STEP 2 · RULES OF ORIGIN & PREFERENTIAL DUTY

MFN Gateway → Origin Proof Mechanism → TCA Preference

Origin determines whether the exporter pays duty at all. Veritariff's origin engine runs a two-stage check: first the MFN gateway (is there even a duty to save?), then the proof mechanism (how do you prove UK origin to claim zero duty under the TCA?)

Step 2.1 — MFN Gateway (Is There a Duty to Save?)

The first question the system asks is: what is the Most Favoured Nation (MFN) duty rate for this commodity code when exported to the EU?

❓ **Is the MFN duty rate for this commodity code 0%?**

✅ **YES** → Zero duty applies regardless of origin. No need to apply for preferential tariff treatment. Skip to Step 3 (Sanctions Check). Note: Veritariff still recommends recording TCA preference for audit trail completeness.

❌ **NO** → A duty saving is available. Proceed to Step 2.2 — prove UK origin to claim 0% preferential rate under the UK-EU TCA.

💡 For most Chapter 72 steel products, the EU MFN rate = 0%. However, certain processed Chapter 73 articles carry MFN rates of 2.7%–6%. Always check per specific 10-digit code via EU TARIC.

Step 2.2 — How Do You Prove Origin?

There are three valid methods to prove origin and claim the UK-EU TCA 0% preferential duty rate. Veritariff presents the user with the appropriate route based on consignment value.

Method 1 — Statement on Origin (Primary Method)

The exporter makes a declaration on a commercial document. The applicable threshold determines what reference number is required.

```
INPUT: consignment_total_value  (converted to EUR)

IF consignment_value_EUR <= 6000  (approx £5,400):
  → ANY exporter may make a Statement on Origin
  → No pre-registration required
  → Place standard TCA Annex ORIG-4 declaration text on:
       - Commercial Invoice, OR
       - Packing List, OR
       - Delivery Note
  → (The document must contain enough detail to identify the goods)
```

```
IF consignment_value_EUR > 6000  (above £5,400):
  → Exporter MUST hold a valid registered exporter reference number
  →
     IF exporter country == 'GB'  (UK exporter):
       → Use EORI Number  (GB + 12 digits)
       → Include EORI number on the Statement on Origin
     IF exporter country IN EU_member_states:
       → Use REX Number  (Registered Exporter System)
       → Apply for REX via national customs authority
```

❓ **Does the exporter have a Statement on Origin (or can they make one)?**

✅ **YES** → Upload the statement (or Veritariff auto-generates it on the commercial invoice). Zero duty rate applies. Proceed to Step 2.3.

❌ **NO** → The exporter can make an origin declaration on a commercial document containing sufficient identifying detail. OR proceed to Method 2 (Importer's Knowledge).

Method 2 — Importer's Knowledge

The German importer may claim preferential tariff treatment based on their own knowledge of the origin of the goods, without requiring a Statement on Origin from the UK exporter.

⚠️ Veritariff must flag to the user: under Importer's Knowledge, the German importer bears full customs liability if their knowledge is subsequently found to be incorrect. Require an importer acknowledgment checkbox before enabling this route.

Method 3 — Other Valid Proofs (Non-TCA Agreements)

For goods not covered by the UK-EU TCA (e.g. third-country goods in transit), non-preferential rules of origin apply. The system must determine the applicable commercial policy measures.

Step 2.3 — What If No Preferential Agreement Applies?

If the goods are not subject to the UK-EU TCA, non-preferential rules of origin apply. The MFN rate applies and the following commercial policy measures may be triggered:

⚠️ Non-preferential origin is used for: MFN treatment, anti-dumping duties, countervailing duties, trade embargoes, safeguarding measures, quantitative restrictions, and tariff quotas.

1. Anti-dumping duties
2. Countervailing duties
3. Trade embargoes
4. Safeguarding measures
5. Quantitative restrictions
6. Tariff quotas

Step 2.4 — Origin Proof Documents Collected

Statement on Origin	TCA Annex ORIG-4 text placed on invoice/packing list/delivery note
EORI Number (UK exporters)	GB + 12-digit EORI — included on Statement for consignments > £5,400
REX Number (EU exporters)	Registered Exporter number — required for EU-side exporters > EUR 6,000
Importer's Knowledge Record	Documented commercial evidence held by German importer
Supplier Declaration	Evidence of originating status of input materials

REG REF	UK-EU TCA Annex ORIG-4 (Statement on Origin text) TCA Art. ORIG.19 (Importer's Knowledge) TCA Art. ORIG.17 (REX — Registered Exporter System) UK Gov Guidance: Claiming preferential tariffs in UK-EU trade EU TARIC Database — MFN rates by CN code

STEP 3 · SANCTIONS COMPLIANCE ENGINE

Non-Russian Origin Verification via Mill Test Certificate

UK and EU sanctions prohibit trade in iron and steel products of Russian origin. Veritariff's sanctions engine performs a three-layer verification before any shipment is cleared. The Mill Test Certificate (MTC) is the primary instrument for proving non-Russian origin.

Step 3.1 — Origin Country Screening

? Is the declared country of origin Russia (RU)?

✅ **YES** → BLOCKED — Refuse shipment creation. Log compliance event. Notify Compliance Officer. Consider SAR obligation.

❌ **NO** → Proceed with risk-tiered screening below.

```
IF declared_origin == 'RU':
  → STATUS: BLOCKED — do not proceed

IF declared_origin IN ['BY','KZ','AM','GE']  [indirect routing risk]:
  → STATUS: ELEVATED_RISK
  → MANDATORY: Steps 3.2 + 3.3 (MTC + Chain of Custody)
  → REQUIRE: Country of MELT AND POUR certificate (not just processing country)

IF declared_origin IN ['UA','MD']  [transshipment risk]:
  → STATUS: STANDARD_ENHANCED
  → REQUIRE: MTC analysis (Step 3.2)

IF declared_origin IN ['GB','DE','FR','PL','TR','KR','CN'...]:
  → STATUS: STANDARD_REVIEW
  → RECOMMEND: MTC spot-check (Step 3.2)

Screen ALL parties against sanctions lists:
  → UK OFSI Consolidated List
  → UN Consolidated List
  → OFAC SDN List
  → EU Consolidated Sanctions List
```

Step 3.2 — Mill Test Certificate (MTC) Verification

The MTC is issued by the steel mill and certifies the chemical composition, mechanical properties, and origin of the steel. It is the key document for proving non-Russian origin.

? Has the exporter uploaded a valid Mill Test Certificate (MTC)?

✅ **YES** → Parse and validate the MTC against the mandatory fields below. Proceed to 3.2.1.

❌ **NO** → HOLD shipment. Request MTC from supplier. Do not proceed until MTC is received and verified.

Step 3.2.1 — Mandatory MTC Fields

STEP 3.2 **MTC Parsing & Validation Logic**

LOGIC

```
EXTRACT AND VALIDATE all of the following from uploaded MTC:

  mtc.steel_mill_name        → screen against OFSI / EU sanctions lists
  mtc.melt_location          → must NOT be 'RU' or sanctioned region
  mtc.pour_location          → same as melt, or document transformation
  mtc.heat_number            → unique batch ID for traceability
  mtc.chemical_composition   → cross-check against Module 1 classification
  mtc.mechanical_properties  → yield strength, tensile strength, elongation
  mtc.product_standard       → EN 10025, ASTM A36, BS EN 10210, etc.
  mtc.certificate_type       → EN 10204 Type 3.1 or 3.2 (preferred)
  mtc.issue_date             → must be within 12 months of shipment date

IF mtc.melt_location == 'RU'  OR  mill_name IN sanctioned_list:
  → STATUS: BLOCKED

IF any mandatory field is missing OR mtc.issue_date > 12 months:
  → STATUS: HOLD — request updated MTC from supplier

IF mtc.chemical_composition differs from Step 1 output by > 0.05%:
  → FLAG DISCREPANCY — require compliance review

IF all checks PASS:
  → STATUS: CLEARED
  → STORE: SHA-256 hash of MTC PDF in immutable Veritariff audit log
```

DATA INPUTS

MTC Document	PDF upload (OCR parsed) or structured XML/JSON feed
Steel Mill Name	Full legal name of the producing mill
Country of Melt	ISO country code where iron ore/scrap was smelted
Heat / Cast Number	Unique production batch identifier for traceability
Chemical Composition	Element % breakdown: C, Si, Mn, P, S, Cr, Ni, Mo, etc.
Product Standard	EN/ISO/ASTM standard certification reference
Issue Date	Date MTC was issued — must be within 12 months
Certificate Type	EN 10204 Type (2.1, 2.2, 3.1 or 3.2)

DOC OUTPUT MTC Verification Report | Sanctions Clearance Certificate | SHA-256 document hash stored in Veritariff audit ledger

REG REF UK Russia (Sanctions)(EU Exit) Regs SI 2019/855 as amended SI 2023/440 | EU Reg 833/2014 Art.3g | DIN EN 10204 (MTC Standard) | OFSI Guidance

Step 3.3 — Chain of Custody (Chapter 73 Articles)

🚫 CRITICAL: Steel of Russian melt origin remains prohibited even if subsequently processed in a third country (e.g. Turkey). Substantial transformation does NOT cure sanctions taint under UK/EU law.

```
FOR CHAPTER 73 ARTICLES (processed steel products):
  → REQUIRE: Full upstream MTC chain:
       Tier 1: Raw steel mill (melt + pour)  →  MTC
       Tier 2: Rolling / processing facility  →  MTC
       Tier 3: UK exporter inventory          →  EORI + stock ref

  IF chain breaks at any tier (supplier cannot provide MTC):
    → REJECT shipment
    → OR: obtain Commodity Trade Surveillance documentation

  Veritariff auto-compiles a Chain of Custody Log:
    Entry 1: Mill name | country | date | heat number
    Entry 2: Processor name | country | date | MTC ref
    Entry 3: UK exporter EORI | date | inventory stock ref
```

REG REF	OFSI Guidance on Russian Steel Sanctions (Feb 2023 update) EU Commission FAQ on Regulation 833/2014 Art.3g HM Treasury OFSI Compliance Guidance — UK financial sanctions

STEP 4 · GERMAN CUSTOMS (ZOLL) IMPORT GATEWAY

EU Import — Surveillance, ATLAS Declaration, EUSt

Upon arrival at a German entry point (Hamburg, Bremen, Duisburg), goods must clear the Bundeszollverwaltung under the EU Union Customs Code. Veritariff pre-populates the German importer's requirements and flags any surveillance document obligations.

Step 4.1 — Steel Safeguard & Surveillance Check (SIGL)

? Is the commodity code subject to EU steel safeguard measures (Regulation 2019/159)?

✅ **YES** → Check TRQ quota availability. If quota remaining → UK TRQ route, no import licence needed. Generate Surveillance Document reference for German importer via BAFA/SIGL portal.

❌ **NO** → No safeguard applies. Proceed to Step 4.2 (ATLAS import declaration).

```
IF commodity_code IN EU_steel_safeguard_products AND origin == 'GB':
  IF TRQ_quota_remaining > 0:
    → UK dedicated TRQ allocation applies
    → NO import licence required
    → GENERATE: Surveillance Document reference for German importer
  IF TRQ_quota_exhausted:
    → ALERT: 25% safeguard duty will apply
    → NOTIFY: importer urgently — commercial decision required

IF Surveillance Regulation 2015/478 triggered:
  → German importer must obtain SIGL Surveillance Document via BAFA
  → MUST be obtained BEFORE shipment departs UK
  IF SIGL document absent → BLOCK shipment creation
```

Step 4.2 — German Import Declaration (ATLAS / H1)

STEP 4.2 — **German ATLAS H1 Import Declaration**

LOGIC

```
German importer files H1 declaration in ATLAS

IF origin == 'GB' AND RoO_proof_valid == TRUE:
  → duty_rate = 0%  (UK-EU TCA preferential)

IF origin == 'GB' AND RoO_proof_valid == FALSE:
  → duty_rate = EU_MFN_rate  (from EU TARIC per CN code)
  → Most Chapter 72 MFN = 0%; Ch.73 articles may be 2.7%-6%

Einfuhrumsatzsteuer (EUSt) — German Import VAT:
  EUSt = 19% x (CIF customs value + customs duty)
  IF importer holds valid DE VAT registration:
    → EUSt deductible as input VAT on German VAT return
```

DATA INPUTS	**German Importer EORI**	DE + 9-digit VAT number format
	DE Customs Office Code	ATLAS office code (Hamburg: 0313000)
	Customs Value (EUR)	CIF value converted to EUR
	Duty Rate	0% (TCA) or MFN rate from EU TARIC
	German VAT Number (USTID)	DE + 9-digit USTID for EUSt relief
	CBAM Declarant ID	CBAM Registry number (if CBAM triggered)
DOC OUTPUT	Pre-populated German H1 Import Declaration XML | ATLAS MRN for German importer	
REG REF	EU UCC Reg 952/2013 | German Zollverwaltungsgesetz | ATLAS System v4.0 | EU Safeguard Reg 2019/159 | EU TARIC	

STEP 5 · **EU CBAM ENGINE**

Carbon Border Adjustment Mechanism — Emissions Data

EU Regulation 2023/956 (CBAM) covers steel goods. During the transitional phase (Oct 2023 – Dec 2025), German importers report embedded carbon quarterly. From Jan 2026, they must purchase CBAM certificates. Veritariff generates the emissions data pack for the UK exporter to transmit to the German importer.

Step 5.1 — CBAM Applicability Gate

❓ **Is the commodity code in CBAM Annex I (steel in-scope goods list)?**

✅ **YES** → CBAM applicable — trigger emissions data collection (Step 5.2). Determine phase: Transitional (reporting only, to Dec 2025) or Definitive (certificate purchase, from Jan 2026).

❌ **NO** → CBAM not applicable for this commodity code. Proceed to Step 6 (Barrister's Bundle).

💡 Principal CBAM steel codes: Chapter 72 headings 7206–7229, select Chapter 73 headings 7301–7309. Always verify specific subheadings against current CBAM Annex I.

Step 5.2 — Embedded Emissions Data Collection

STEP 5.2 **Specific Embedded Emissions (SEE) Calculation**

LOGIC

```
SEE (tCO2e/tonne) = Direct_Emissions + Indirect_Emissions

IF production_route == 'BF-BOF':
  direct   = (hot_metal_carbon * emission_factor)
           + limestone_decomposition_CO2
           + (gas_GJ * 56.1 tCO2/TJ)
  indirect = electricity_kWh * GB_grid_factor

IF production_route == 'EAF':
  direct   = electrode_kg * 3.636 tCO2/t_graphite
           + (gas_GJ * 56.1 tCO2/TJ)
  indirect = electricity_kWh * 0.233 tCO2/MWh  [UK DESNZ 2023]

IF production_route == 'DRI-EAF':
  direct = DRI_CO2 + EAF_direct

IF SEE data unavailable:
  → USE EU Commission default values
  → WARN user: defaults are conservative = higher CBAM cost for importer

IF UK ETS carbon price was paid:
  → Calculate offset per Art.9 CBAM Reg 2023/956
  → offset = MIN(UK_ETS_equivalent, EU_ETS_weekly_price)
```

DATA INPUTS	**Production Route**	BF-BOF | EAF | DRI-EAF
	Direct CO2 (tCO2e/t)	From furnace operations and fuel combustion
	Indirect CO2 (tCO2e/t)	From purchased electricity
	Electricity (kWh/t)	Per tonne of steel produced
	UK ETS Carbon Price (£/t)	Actual carbon cost paid — for CBAM offset calc
	EU ETS Weekly Price (EUR/t)	ICE Endex EUA price for certificate cost calc
DOC OUTPUT	CBAM Embedded Emissions Certificate | Quarterly Report XML (EU CBAM Registry format) | Carbon Offset Worksheet	
REG REF	EU CBAM Reg 2023/956 Arts.6–9 | EU Implementing Reg 2023/1773 Annex II | DESNZ Grid Emission Factors 2023	

STEP 6 · HMRC CDS EXPORT DECLARATION

Customs Declaration Service — UK Export Data Elements

All UK steel exports must be lodged as a full export declaration in HMRC's Customs Declaration Service (CDS) before the goods reach the UK frontier. Veritariff auto-populates all Data Elements (DEs) from the shipment record.

Step 6.1 — Declaration Type Gate

? Is the export value >= £1,500?

✅ **YES** → Full Export Declaration required (procedure type 'EX'). Pre-lodge before goods arrive at frontier. Proceed to Step 6.2.

❌ **NO** → Simplified Declaration may be used. Note: if commodity code is a sanctioned steel grade, full declaration is mandatory regardless of value.

Step 6.2 — Core Data Elements

STEP 6.2	CDS Mandatory Data Elements	
HEADER DES	**DE 1/1 — Declaration Type**	EX (Export), CO (Outward Processing)
	DE 1/2 — Additional Type	A (Standard), D (Pre-lodged)
	DE 3/1 — Exporter EORI	GB + 12-digit EORI
	DE 5/8 — Country of Destination	DE (Germany)
	DE 5/14 — Country of Dispatch	GB
GOODS DES	**DE 6/8 — Commodity Code**	10-digit UK Tariff code (from Step 1)
	DE 6/1 — Net Mass (kg)	Weight excluding packaging
	DE 6/2 — Supplementary Units	Tonnes where required by tariff
	DE 6/5 — Gross Mass (kg)	Total weight including packaging
	DE 1/10 — Procedure Code	10 00 (permanent export), 21 00 (OP), 31 00 (re-export)
VALUATION DES	**DE 4/1 — Delivery Terms**	Incoterms 2020 code (FOB, CIF, DAP…)
	DE 4/14 — Invoice Currency	ISO 4217 code
	DE 4/16 — Valuation Method	1 (transaction value) — preferred
VALIDATION	EORI: validate via HMRC EORI checker API before lodgement. 10-digit code: validate against UK Trade Tariff API on every declaration. Non-GBP invoices: auto-convert using HMRC weekly exchange rate.	
REG REF	UK Tariff Vol 3: Data Elements for CDS Export | HMRC Notice 252 (Customs Valuation) | WTO Valuation Agreement	

Step 6.3 — Safety & Security Exit Summary

```
IF exit_via == 'RoRo'             → submit EXS at least 2 hours before departure
IF exit_via == 'container_sea'    → submit EXS at least 24 hours before loading
IF exit_via == 'air_freight'      → submit EXS at least 30 minutes before departure
IF AEO_status IN ['AEOC','AEOS'] → EXS waived — suppress generation, flag exemption
```

STEP 7 · THE BARRISTER'S BUNDLE

Complete Document Checklist & Pre-Shipment Release Gate

The Barrister's Bundle is Veritariff's master compliance pack — assembled automatically as each preceding step completes. No shipment is released until every mandatory document is validated and present. This is the audit trail that defends the transaction in any legal, customs, or enforcement proceeding.

#	Document	Step	Source	Condition
1	**Commercial Invoice**	Step 6	Exporter	MANDATORY — always
2	**Packing List**	Step 6	Exporter / Warehouse	MANDATORY — always
3	**Bill of Lading / CMR**	Step 6	Freight Forwarder	MANDATORY — transport mode dependent
4	**UK CDS Export Declaration (MRN)**	Step 6	Veritariff / CDS API	MANDATORY — always
5	**Exit Summary Declaration (EXS)**	Step 6.3	Veritariff / CDS	MANDATORY unless AEO-exempt
6	**Mill Test Certificate (MTC)**	Step 3.2	Steel Producer	MANDATORY for all steel
7	**Sanctions Screening Certificate**	Step 3.1	Veritariff (auto)	MANDATORY — always
8	**Chain of Custody Log**	Step 3.3	Veritariff (auto)	MANDATORY for Chapter 73 articles
9	**Statement on Origin / EORI Decl.**	Step 2.2	Veritariff (auto)	IF TCA preference claimed
10	**Supplier Declaration (inputs)**	Step 2.4	Exporter's suppliers	IF TCA PSR compliance needed
11	**CBAM Emissions Certificate**	Step 5.2	Veritariff (auto)	IF CBAM applicable (Ch.72/73)
12	**CBAM Quarterly Report XML**	Step 5.2	Veritariff (auto)	IF in CBAM transitional phase
13	**SIGL Surveillance Document**	Step 4.1	BAFA Germany	IF EU safeguard quota triggered
14	**AEO Certificate**	External	HMRC (pre-issued)	IF AEO authorised exporter

Step 7.1 — Pre-Shipment Release Gate

```
// ALL conditions must pass before shipment_status = 'CLEARED_FOR_EXPORT'

MANDATORY = ['commercial_invoice','packing_list','cds_mrn',
             'mtc_verified','sanctions_cleared']

FOR EACH doc IN MANDATORY:
  IF doc.status != 'VALIDATED':
    → BLOCK release
    → DISPLAY: doc.name + action_required

IF tca_preference_claimed AND origin_declaration.status != 'SIGNED':
  → BLOCK release

IF cbam_applicable AND cbam_data.status != 'COMPLETE':
  → BLOCK release

IF ALL checks PASS:
  → shipment_status = 'CLEARED_FOR_EXPORT'
  → GENERATE: Veritariff Clearance Certificate (timestamped + SHA-256 signed)
  → BUNDLE: ZIP all docs → S3 storage with AES-256 encryption
  → NOTIFY: exporter + freight forwarder + German importer
```

APPENDIX A — Regulatory Reference Index

UK Trade Tariff API	api.trade-tariff.service.gov.uk
UK-EU TCA Rules of Origin	TCA Annex ORIG-1 through ORIG-7 (legislation.gov.uk)
UK Sanctions (Russia)	SI 2019/855 as amended by SI 2023/440
OFSI Guidance	gov.uk/government/organisations/office-of-financial-sanctions-implementation
EU CBAM Regulation	EU Reg 2023/956 — eur-lex.europa.eu
EU CBAM Transitional	EU Implementing Reg 2023/1773
EU Union Customs Code	EU Reg 952/2013
EU Steel Safeguard	EU Reg 2019/159 + Council Implementing Reg 2023/35
EU TARIC	ec.europa.eu/taxation_customs/dds2/taric
German ATLAS	Bundeszollverwaltung — zoll.de
German BAFA (SIGL)	Bundesamt für Wirtschaft und Ausfuhrkontrolle — bafa.de
EN 10204 (MTC Standard)	DIN EN 10204:2004 — Types 2.1, 2.2, 3.1, 3.2
HMRC Notice 252	gov.uk/guidance/notice-252-valuation
Open Government Licence	nationalarchives.gov.uk/doc/open-government-licence/version/3/

APPENDIX B — API Integration Map

UK Trade Tariff API	api.trade-tariff.service.gov.uk/#what-can-you-do-with-the-apis
HMRC EORI Checker	api.service.hmrc.gov.uk/customs/eori/GB{number}
OFSI Sanctions List	gov.uk/government/publications/financial-sanctions-consolidated-list-of-targets
EU EORI Validation	ec.europa.eu/taxation_customs/dds2/eos/eori_validation
EU TARIC API	ec.europa.eu/taxation_customs/dds2/taric/taric_consultation
EU Financial Sanctions	data.europa.eu/euodp/en/data/dataset/consolidated-list-of-persons
EU CBAM Registry	cbam.ec.europa.eu
ICE Endex ETS Price	theice.com/marketdata (EU ETS EUA price)
UK ETS Auction Results	gov.uk/guidance/uk-ets-auction-results
HMRC Exchange Rates	gov.uk/government/collections/exchange-rates-for-customs

APPENDIX C — Veritariff Product Architecture Notes

Based on the veritariffai.co product vision, the following platform modules map to this functional specification:

Pre-Shipment Margin Simulation	Step 1 (Classification) + Step 2 (Origin / MFN duty rate) → landed cost engine
HS Code AI	Step 1.1–1.2 (Steel Logic Tree) → chemical composition + physical form classifier
Rules of Origin Engine	Step 2.1–2.4 → MFN gateway, TCA threshold, EORI/REX/Statement on Origin
Compliance Shield / Sanctions Scan	Step 3 → OFSI/OFAC/EU list screening + MTC verification engine
Digital Product Passport	Step 5 (CBAM) + Step 2 (RoO) → embedded CO2 data + origin certificate
AEO-in-a-Box	Step 7 (Barrister's Bundle) → automated audit trail, document vault, clearance cert
Incoterm Modelling	Step 6 (CDS DE 4/1 delivery terms) → CIF/FOB/DAP valuation impact
FX Integration	Step 6 (DE 4/14) → HMRC weekly exchange rate conversion